Program : **Diploma in Cloud Computing and Big Data**	
Course Code : **5279C**	Course Title: **Data Mining Practices Lab**
Semester : **5**	Credits: **1.5**
Course Category: **Programme Elective Course**	
Periods per week: **3 (L: 0 T: 0 P: 3)**	Periods per semester:**45**

Course Objectives:

- Understand data and data preprocessing
- Gain knowledge about data mining techniques

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Programming Techniques		Programming in python lab	3

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Implement Matrix operations and linear algebra on matrices using python	9	Applying
CO2	Practice loading and Preprocessing data using python	9	Applying
CO3	Demonstrate Visualization of data using python	9	Applying
CO4	Use data mining in python	12	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	2					
CO3	3	2					
CO4	3	2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	**Implement Matrix operations and linear algebra on matrices using Python**		
M1.01	Implement multi-dimensional arrays and find its shape and dimension	2	Applying
M1.02	Reshape and flatten data in the array	1	Applying
M1.03	Apply indexing and slicing on array	2	Applying
M1.04	Practice Dot and matrix product of two arrays	2	Applying
M1.05	Compute the multiplicative inverse of a matrix	2	Applying
CO2	**Practice loading and Preprocessing data using Python**		
M2.01	Loading data from CSV file	2	Applying

M2.02	Compute the basic statistics of given data - shape, no. of columns, mean	2	Applying
M2.03	Practice Preprocessing techniques for missing data, redundancy, null value, and noisy data	3	Applying
M2.04	Splitting a data frame on values of categorical variables	2	Applying
	Lab Test – I	3	
CO3	**Demonstrate Visualization of data using Python**		
M3.01	Practice data split for training & test and visualize the data	2	Applying
M3.02	Perform various graphical representation of data- Line Plot, Scatter Plot, Box plot, Point plot, Count plot, Violin plot, Swarm plot, and Bar plot	7	Applying
CO4	**Use data mining in python**		
M4.01	Practice web scraping	5	Applying
M4.02	Open ended projects- Design and develop python programs to analyze social media data or e-commerce data	7	Applying
	Lab Test – II	3	

Text / Reference

T/R	**Book Title/Author**
T1	1. J. Han, M. Kamber, “Data Mining Concepts and Techniques”, Morgan Kaufmann.
T2	2. M. Kantardzic, “Data mining: Concepts, models, methods and algorithms, John Wiley &Sons Inc.
R3	3. M. Dunham, “Data Mining: Introductory and Advanced Topics”, Pearson Education

Online Resources

Sl.No	Website Link
1	https://www.dataquest.io/blog/sci-kit-learn-tutorial/
2	https://www.ibm.com/support/knowledgecenter/en/SS3RA7_sub/modeler_tutorial_ddita/modeler_tuto rial_ddita-gentopic1.html
3	https://archive.ics.uci.edu/ml/datasets.php